A4.1 Evolution and speciation

NAME

DATE

How to use this sheet: go through the slides with it, then close them and write from memory. Use your own words and add page numbers from the book (pp. 138–150) where you want more detail.

Guiding questions. What evidence supports evolution? What do analogous and homologous structures show about what organisms share and how they differ?

A4.1.1 Evolution

Define evolution. How did the ideas of Lamarck and Darwin differ?

A4.1.2 Sequence evidence

Why do close relatives have more similar sequences? Name one gene that is widely shared.

A4.1.3 Selective breeding

What does selective breeding show, and what does it not prove? Give two examples.

A4.1.4–5 Homologous and analogous structures

	HOMOLOGOUS	ANALOGOUS
Basic structure		
Function		
Origin		
Example		

Sketch and label the bones of a pentadactyl limb.

A4.1.6–7 Speciation

Define speciation. Explain what reproductive isolation and differential selection each contribute, using the example of bonobos and chimpanzees.

Before you finish

Tick the ones you can explain without notes. Circle anything to ask about.

- ☐ Definition of evolution
- ☐ Lamarck and Darwin
- ☐ Sequence evidence
- ☐ Selective breeding
- ☐ Pentadactyl limb
- ☐ Analogous structures
- ☐ Speciation
- ☐ Bonobos and chimpanzees

What the Cell. Original material written for this course.